

Sriharsha Meduri

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WORK EXPERIENCE

Head of Engineering - OviGuide

03/2026 - Present

- Architected the **admission prediction engine** achieving high accuracy across **50,000+** JoSAA college-branch combinations using historical cutoff data (2019–2025).
- Developed the real-time **Counselling Copilot** that analyzes seat allotments every round and provides **Freeze/Float/Slide** recommendations with risk scores.
- Built the **Truth Engine** aggregating placement statistics, Reddit insights, and NIRF rankings to generate optimized choice lists and transparent college intelligence.

Technologies: Python, React, Next.js, FastAPI, Pandas, Scikit-learn

CTO & Co-Founder - QuickPe

02/2026 - Present

- Designed and implemented an **offline-first biometric payment system** supporting fingerprint authentication with **ECDSA P-256** signing and secure enclave storage.
- Built the offline transaction queuing mechanism with strict velocity limits and automatic cloud sync.
- Led end-to-end architecture including merchant-consumer flows, anomaly-based **fraud detection**, and hardware-backed private key management.

Technologies: React, Next.js, Node.js, ECDSA Cryptography, Biometric APIs

Founding Engineer - Oviqo

01/2026 - Present

- Designed and implemented the core **personalized video generation engine** using fine-tuned diffusion models and TTS pipelines to create unique teaching videos adapted to each student's learning style.
- Built the full backend architecture for material ingestion, **concept extraction**, knowledge graph construction, and adaptive learning path generation.

Technologies: Python, PyTorch, React, Next.js, FastAPI, Diffusion Models

Research Intern (Machine Learning) - IIM Shillong

09/2025 - Present

- Built a latency-optimized financial complaint classification system using **BiLSTM** with domain-adaptive **GloVe embeddings** trained on banking-specific corpus.
- Achieved **85.3% accuracy** while reducing inference latency by **16x** (1.1ms vs 17.6ms) compared to DistilBERT baseline.
- Optimized model size and inference pipeline making it suitable for real-time deployment on low-resource banking servers.

Machine Learning Intern - iMerit Technology Services Pvt. Ltd.

12/2025 - 03/2026

- Developing a post-hoc **Out-of-Distribution (OOD)** detection pipeline for semantic segmentation in autonomous driving, evaluated on KITTI, Fishyscapes, and nuScenes datasets.
- Implemented **MSP**, **ODIN**, and **Energy-score** methods to detect anomalous objects and edge cases in driving scenes.
- Designed a failure mode analysis dashboard for the improvement of perception systems.

PROJECTS

VisionGuardAI - Real-time Container Damage Detection

[Github](#) || [Demo](#)

- Built an end-to-end **edge AI** system using fine-tuned **YOLOv5** on Jetson Nano for real-time detection of five container damage types (Hole, Dent, Rust, Deframe, Mini-Dent).
- Achieved **98.7% mAP** with inference latency under **50ms** and real-time WebSocket streaming to the operator dashboard.
- Implemented severity scoring, bounding box confidence overlays, multi-camera support, activity logging, and automated CSV report export.

Technologies: YOLOv5, PyTorch, WebSocket, React, Next.js, Jetson Nano, Tailwind CSS

- Led frontend development of the Andhra University’s 100-year **centenary celebrations website** (Shatabdi Mahotsav 1926–2026).
 - Implemented interactive features including event calendar, live countdown timers, global time zone display, and dynamic registration system.
 - Designed fully responsive UI with rich content sections on university history and achievements.
- Technologies:** React, Next.js, Tailwind CSS, JavaScript, Framer motion

Department Day Event Management System

Github || Demo

- Developed a full-stack-ready responsive web application for managing college Department Day events with browsing, registration, and admin workflows.
 - Implemented dynamic filtering, search, sorting, grid/list views, and multi-step registration with form validation and image upload preview.
- Technologies:** React, Next.js, JavaScript, Tailwind CSS, Framer Motion

LiveGuard AI - Multi-Modal Forensics

Github || Demo

- Developed a multi-modal forensics system combining **CLIP-based Deepfake Detection** and **XCLIP** for synthetic media analysis with explainable risk scoring.
 - Deployed scalable analysis modules using **FastAPI + Docker** backend with responsive **React** frontend.
 - Implemented aggregated risk scoring engine fusing multi-modal signals for transparent deepfake and synthetic media assessment.
- Technologies:** Python, FastAPI, React, Docker, PyTorch, XCLIP

PhishingLens - Phishing Detection System

Github || Demo

- Built a production-grade phishing detection system achieving **96% accuracy** by fusing OTX, GNN, and Distil-BERT signals through consensus voting.
 - Reduced false positives by **40%** and deployed a secure Chrome extension used by over 500 users.
 - Architected Dockerized microservices backend with Prometheus monitoring and real-time threat intelligence pipeline.
- Technologies:** Python, Docker, GNN, Transformers, React, AWS

ACHIEVEMENTS

- **Winner in National Cybersecurity Hackathon** organised by Ministry of Electronics and IT.
- **Round 2 Qualifier** in **DP World Hackathon 2026** at BITS Pilani.
- **Round 3 Qualifier** in **MumbaiHacks 2025**, competing among top national teams.
- **Qualified Round 1 (Internal)** in **Smart India Hackathon (SIH) 2025**.
- **Co-Lead** at **GDSC Machine Learning Chapter**, organizing workshops and CTFs for **200+ students**.
- **Published Research** on Deep Learning for Text Classification (Under review).

EDUCATION

08/2023 - 04/2027

B.Tech in Information Technology at **Andhra University** (AUCE)
Coursework: Data Structures & Algorithms, DBMS, OS, Computer Networks.

CGPA: 8.0/10

SKILLS

Languages:

Python, JavaScript, TypeScript, Java, C, SQL.

Web Dev:

React, Next.js, FastAPI, Django, Tailwind CSS, HTML, Node.js, Microservices, MySQL.

ML / AI:

PyTorch, Scikit-learn, Transformers, GNN, YOLOv5/v8, Langchain, TensorFlow.

DevOps:

Docker, Kubernetes, AWS, GCP, REST APIs, CI/CD, GitHub Actions.

Tools:

Git, Github, Figma, Wix, Notion, OAuth, Cursor, Postman.

LANGUAGES

Telugu, English, Hindi